

Varshini Akula

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Software engineer with two years of production experience at JPMorgan Chase building full-stack platforms, Java and Spring Boot services, and high-volume data pipelines under high-availability SLAs; strengths in distributed systems, REST APIs, AWS, and shipping systems end to end.

EDUCATION

Arizona State University

Master of Science in Computer Science, GPA 4.00/4.00

Tempe, AZ, USA

Aug 2025 - May 2027

Jawaharlal Nehru Technological University

Bachelor of Technology in Computer Science and Engineering (Data Science), 9.49/10 CGPA

Hyderabad, India

Dec 2020 - May 2024

PROFESSIONAL EXPERIENCE

JPMorgan Chase, Hyderabad, India: Software Development Engineer I

Jul 2024 - Aug 2025

- Architected and shipped a **three-service platform on AWS** (Rasa NLU engine, **Python** action service, **TypeScript** front end) that replaced **320+ hours of manual onboarding calls per quarter** with an automated flow, reaching production in under **6 months**.
- Built the metadata capture and document-processing pipeline for a system ingesting **40M+ documents per day**, persisting extracted variables from **AWS S3** to **MongoDB** through **CRUD APIs**.
- Improved production reliability with **structured logging, distributed tracing, and health-check endpoints** under a **99% uptime SLA**; named **subject matter expert for NLP and AI** after launch.

JPMorgan Chase, Hyderabad, India: Software Engineering Program Attachment Intern

Jan 2024 - Jun 2024

- Designed and built a **React and TypeScript** incident console with a chatbot grounded in historical **P1-P4** resolution data, cutting mean incident resolution time by **60%**.
- Built a runbook assistant over **document embeddings and vector search** with a rule-based fallback, onboarding engineers onto **Consumer & Community Banking** applications; migrated a **.NET** test suite to **Java**.

JPMorgan Chase, Hyderabad, India: Software Engineering Intern

May 2023 - Jul 2023

- Integrated a virtual assistant into an **Apache Kafka** publisher/consumer log-streaming platform serving **70+ downstream applications** and upgraded versioning across **Java and Spring Boot** services with **JUnit and Mockito** coverage.

PROJECTS

FlowMind: Visual AI-Assistant Builder

Summer 2026

- Built FlowMind (**Next.js 15, React 19, TypeScript, Supabase/Postgres**), a visual AI-assistant builder with a typed node-graph execution engine, **pgvector/HNSW** retrieval with **BM25** fallback, **multi-tenant row-level security** verified by an automated cross-tenant test suite, and an offline evaluation harness with **CI** (GitHub Actions).

Construction Project Intelligence

Spring 2026

- Built a full-stack **FastAPI and React** document-intelligence platform with **role-based access control (RBAC)**, **PostgreSQL-backed indexing**, RFI classification, and real-time analytics dashboards, cutting manual lookup time by **95%**.

ClickLess AI, Conversational Travel Agent

Spring 2026

- Built a conversational travel agent using the **OpenAI API** for intent handling, **Playwright** for browser automation, and a **FastAPI** backend across **5+ integrated services**.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, Bash

Backend: Spring Boot, FastAPI, REST APIs, Apache Kafka, OAuth, WebSocket, microservices, JUnit, Mockito, pytest

Frontend: React, Next.js, Tailwind CSS, HTML, CSS

Databases & Cloud: MongoDB, PostgreSQL, pgvector, Supabase, SQL, Redis, AWS (S3, EC2, Lambda), Docker, Kubernetes, CI/CD, Git, Linux

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, RAG, vector search, LLM fine-tuning, LLM APIs

Certifications: AWS Certified Cloud Practitioner (2024), DBMS (NPTEL, IIT Kharagpur)

AWARDS & LEADERSHIP

Awards: JPMorgan Code for Good - 1st Place; NASA Space Apps Hackathon - Finalist

Leadership: Engagement Manager, ACM at ASU (4,700+ members, 2025-26); Student Chair, Women's Leadership Conclave 2024, GNITS